

Program : Diploma in Textile Technology	
Course Code : 4061	Course Title: Yarn Manufacture II
Semester : 4	Credits: 4
Course Category: Program core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To impart the knowledge of constructional features and working of Speed frame, Ring frame, doubling machines & advanced spinning systems
- To understand the process involved in the conversion of sliver to yarn
- To estimate the draft, production and efficiency of the machines.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Basic mathematics		Mathematics I, II	1, 2
Basics of mechanical elements in machine		Basic Textile mechanics lab	2
Pre-spinning process		Yarn Manufacture I	3

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the working of Speed frame and Ring frame.	18	Applying
CO2	Describe the process of yarn formation through condensed yarn spinning and other advanced spinning techniques.	15	Understanding
CO3	Discuss the different methods of yarn doubling techniques and reeling of yarn.	10	Understanding
CO4	Illustrate and solve the problems related to draft, twist, angle of yarn pull, traveler lag and determine production of various machines.	15	Applying
	Series Test	2	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	1					
CO2	3						
CO3	3						
CO4	3	3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive levels
CO1	Illustrate the working of Speed frame and Ring frame		
M1.01	Explain the working of simplex machine and the functions of the parts involved.	5	Understanding
M1.02	Discuss the modern developments in speed frame.	4	Understanding
M1.03	Describe the working of Ring frame and the functions of the parts involved and building mechanism of Ring frame.	5	Understanding
M1.04	Identify and analyse the possible Ring frame processing defects and suggest remedies to solve them.	4	Applying

Contents:

Objectives of Speed frame - working of speed frame machine- SKF PK 1500 drafting system - Constructional details of flyer assembly - Function of cots, condenser, false twister, spindle, Principles of winding - Flyer lead and bobbin lead, objectives of builder motion, Objectives of Differential motion, Developments in speed frame - Features of multi motor drive system, Roving stop motion using light barriers, Auto doffing and bobbin transport system.

Objectives of Ring frame - working of Ring frame machine - LR pneumatic top arm drafting system- function of bobbin holder, roving traverse guide, fluted and knurled roller, spacer, cradle, nose bar, roller clearer, pneumafil suction system, Traversing thread guide, ABC ring, separator, top roller cots and shore hardness, aprons-Rings Types - Low crown ring & Anti-wedge ring - Comparison of C-traveller and Elliptical traveler - concept of Traveller number and Ring Flange number ,Running in period of ring, Construction details of Modern spindle, Twisting principle, objectives of building mechanism, working of building mechanism of Ring frame for weft wind, Methods for achieving variable spindle speed in spinning - VFD & VPS. Identify and analyze the possible Ring frame processing defects such as excessive end breaks, slubs, thick and thin places, soft yarn, neps, snarls and suggest remedies to solve them.

CO2	Describe the process of yarn formation through condensed yarn spinning and advanced spinning techniques.		
M2.01	Explain the spinning geometry, mechanism of compact spinning and benefits of compact yarn	3	Understanding
M2.02	Discuss the principle of open end spinning and working of Rotor spinning.	4	Understanding
M2.03	Illustrate the principle of Friction spinning and working of DREF II spinning machine.	4	Understanding
M2.04	Explain the principle of Air jet spinning and working of Murata Air jet spinner.	4	Understanding
	Series Test - I	1	

Contents:

Need for Compacting element in condensed yarn spinning - concept of spinning Geometry - Working of Elite compact spinning system - advantages of compact yarn - Working of SIRO spinning-Different advanced spinning systems - principle of open end spinning system- working of Rotor spinning system - Functions of opening roller, transport tube, Rotor, navel, take up tube - Relation between fibre length and rotor diameter, rotor speed - Types of rotor groove - advantages and limitations of rotor spinning over ring spinning - Principle of friction spinning - working of DREF II friction spinning - Principle of air jet spinning - fasciated yarn - working of Murata Air jet spinning - comparison of yarn structure and properties of ring, rotor and air jet yarn.

CO3	Describe the different mechanism of yarn doubling techniques and reeling of yarn.		
M3.01	Discuss the objectives and the methods of doubling	2	Understanding
M3.02	Illustrate the working of Ring doubler and TFO machine	3	Understanding
M3.03	Discuss the types of fancy yarn and working of fancy doubler.	3	Understanding
M3.04	Describe the objectives of reeling and the concept of bundling and baling.	2	Understanding

Contents

Objects of doubling - dry and wet doubling - English and scotch system of wet doubling - Effect of Twist direction - D/S ratio of doubled yarn - Types of doubled yarn - ply, cord, cable, poplin, sewing thread, voile - Types of doubling machine - working of Ring doubler - Objectives of assembly winder - working of TFO - Merits of TFO over Ring doubler - Define fancy yarn - Types of fancy yarn - slub yarn, marl yarn, corkscrew yarn, loop yarn and snarl yarn - Working of Fancy doubler - Objectives of Reeling - working of Plain reeling machine - Concept of bundling and baling - relationship between number of knots in a bundle and count.

CO4	Illustrate the concept of draft, twist, angle of yarn pull, traveler lag and determine production of speed frame, ring frame, rotor spinning, Ring doubler and TFO machine.		
M4.01	Determine the TPI, actual draft, production and machine efficiency of speed frame	4	Applying
M4.02	Determine the TPI, actual draft, angle of yarn pull, traveller speed, traveller lag, and twist contraction in Ring frame.	4	Applying
M4.03	Determine the production and machine efficiency of Ring frame and Rotor spinning machine.	4	Applying
M4.04	Determine the resultant count, production and efficiency of Ring doubler and TFO machine	3	Applying
	Series Test - II	1	
Contents: Roving hank calculation - TPI and actual draft calculation of speed frame & ring frame - concept of angle of yarn pull - traveler speed - traveller lag - twist contraction and its calculation - production and efficiency of speed frame & ring frame - production & efficiency of Rotor spinning machine - Resultant count - production and efficiency of Ring doubler & TFO machine.			

Text / Reference:

T/R	Book Title/Author
T1	W. Klein, Rieter Manual of Spinning Volume 4-6 , Rieter, 2010
R2	R. V. M. Gowda, New Spinning Systems, NCUTE Publication, New Delhi, 2005
R3	Oxtoby E., “Spun Yarn Technology”, Butterworths, London, 2002.
R4	Dr.A.R.Khare- Elements of Ring Frame and Doubling
R5	Chattopadhyay R, “Advances in Technology of Yarn Production”, NCUTE, IIT Delhi, 2002.

Online resources:

Sl.No	Website Link
1	https://nptel.ac.in
2	www.reiter.com
3	www.laksmimach.com
4	www.suessen.com
5	www.fibre2fashion.com